John (Feng) Huang

Senior AI & Systems Engineer | Manufacturing & Industrial Domain

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EXECUTIVE SUMMARY

20+ years of engineering leadership bridging heavy industry/precision manufacturing (cleanroom FPD, medical devices, heavy machinery) and modern AI/software platforms. Deep first-hand domain expertise in industrial time-series data, MES architecture, and shop-floor operational friction. Proven capability building resilient, production-grade AI Agent workflows, RAG systems, and local LLM microservices - focusing on deterministic business logic and operational value over brittle demo scripts.

CORE COMPETENCIES & TECHNICAL SKILLS

AI & Agent Engineering:	LLM Tool-Use / MCP Workflows, RAG Pipelines, Local LLM Inference (vLLM / Qwen), LiteLLM Gateway, Agent Evaluation & Observability
Industrial & Domain Systems:	MES & EAM Architecture, Industrial Time-Series Data Governance, Process Automation, ISO 13485 Quality Traceability, SAP Integration
Platform & Software:	Python (FastAPI, Asyncio, Data Pipelines), Linux Systems & Automation, Containerization (Docker), SQL/Postgres, Pragmatic Vibe Coding
Collaboration:	Autonomous Execution, Asynchronous Cross-Timezone Collaboration, Technical Documentation, Bridge between Business & Engineering

PROFESSIONAL EXPERIENCE

AI Product Manager / Lead Engineer
2026.05 - 2026.09

Xinghuo Shuzhi (EdTech / AI Platform)

- Delivered full-stack AI platform end-to-end: prototyping, dual knowledge graphs, private Qwen local deployment, and 3 production microservices.
- Engineered fully local LLM stack (Qwen3.6-35B / Qwen3.8-27B / bge-m3) with LiteLLM gateway, guaranteeing zero data leakage outside intranet.
- Built document AI pipeline from 0 to 1: automatic workbook segmentation, RapidOCR, formula parsing (KaTeX), and YOLOv8 pre-labeling workflow.
- Established human-AI architectural division: deterministic algorithms for mastery evaluation, LLMs for qualitative insight generation.

Digital Equipment Manager / EAM Project Lead
2021.06 - 2025.11

Sany Group (Heavy Construction Machinery)

- Orchestrated enterprise EAM rollout across 9 Business Units and 40+ factories, achieving 100% online operational coverage.
- Pioneered LLM+RAG Maintenance AI Assistant in manufacturing, achieving 63% solution adoption rate and reducing reliance on senior specialists.
- Closed data loop between SAP depreciation and MES OEE calculation; governed 300K+ equipment master data records (data cleanliness +40%).
- Built multi-criteria evaluation model for idle asset redeployment (~$3M USD value), driving annual equipment cash expenses down by ~8%.

Process Equipment & Automation Specialist
Earlier Career

CSOT (FPD High-Tech Cleanroom) & Medical Device Manufacturing

- Managed high-precision coating process equipment and MES execution in cleanroom flat panel display (CSOT) production lines.
- Enforced strict process validation, batch record tracking, and quality traceability under rigorous medical device standards (ISO 13485).
- Deeply witnessed the real-world friction of industrial data collection ('collecting everything, utilizing nothing'), shaping a lean approach to digital architecture.

CERTIFICATIONS & LANGUAGES

AI Certifications:	Anthropic Certified (Claude 101, AI Fluency, Agent Skills, Claude Code in Action - Mar 2026)
Languages:	Chinese (Native), English (Professional working proficiency, async/documentation), German (Reading & technical context)